

SAISAB SADHU

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EDUCATION

Indian Institute of Science Education and Research Bhopal (IISER Bhopal) 12/2021 - 04/2026
BS-MS (Integrated) in Data Science and Engineering CPI : 7.58/10

INDUSTRY EXPERIENCE

Lexsi Labs March 2026 – Present
AI Research Scientist *Mumbai, India*

- Research across mechanistic interpretability, machine unlearning, compliance auditing, and safety alignment, spanning most work published, accepted, or under review since joining (see Publications).
- Developing SafeTune, an open-source library for safety-alignment workflows covering guardrail, evaluation, and safety-aware fine-tuning.

MIQ Digital Jan 2026 – March 2026
Senior Analyst Intern, Data and Analytics Team *Bengaluru, India*

- Analyzed large-scale user datasets on Databricks over AWS S3; applied statistical methods and ML models for campaign optimization and attribution modeling.

RESEARCH EXPERIENCE

Bio Medical Data Science Lab, IISER Bhopal Jan 2025 – May 2026
Graduate & Undergraduate Researcher *Bhopal, India*
PI: Dr. Tanmay Basu — Collaborators: Dr. Dwaipayan Roy (IISERB), Dr. Biswajit Patra (IISERB)

- **MS Thesis:** Investigating mechanisms to overcome imperfect context retrieval and resolve knowledge conflicts, both **parametric** and **external** in RAG Frameworks. Developed a dialectical engine that operationalizes formal cross-examination to dynamically assess source credibility based on logical resilience, achieving 77% improvement on FaithEval and 28% on RAMDocs.
- **BS Thesis:** Engineered a hybrid extractive–abstractive summarization pipeline achieving strong performance (53.13 ROUGE-1) on CNN/DailyMail; developed a ModernBERT-based Siamese extractive stage using a scaled adaptive margin triplet loss for optimal candidate ranking.
- **Biomedical NLP:** Developing an end-to-end deep learning framework for automated PICO (Population, Intervention, Comparison, Outcome) extraction from full-text clinical papers in collaboration with ICMR Bhopal to curate annotated datasets of full research articles to accelerate evidence synthesis for systematic reviews.

School of Public Policy, IIT Delhi May 2024 – July 2024
Research Intern *New Delhi, India*
Guide: Dr. Nandana Sengupta; Co-Guide: Prof. Ravinder Kaur, Prof. Sangeeta Kohli

- Analyzed 2000+ faculty profiles (IRINS) to identify a 12% gender differential in negative marking impact; evaluated the socio-economic viability and impact of the 20% supernumerary quota for women at IITs.

CONFERENCE PROCEEDINGS

- [C1] (Best Paper Award) Saisab Sadhu, Dwaipayan Roy, and Tanmay Basu. “[When RAG Disagrees: Detecting Latent Epistemic Conflict via Logit Interactions](#)”. In *ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*.
- [C2] Saisab Sadhu, Dwaipayan Roy, and Tanmay Basu. “[DARE: A Dialectical Framework for Adversarial and Evidence-Aware RAG](#)”. In *European Conference on Information Retrieval (ECIR)*. 2026.
- [C3] Pratinav Seth*, Saisab Sadhu*, Anshul Kaushal*, and Vinay Kumar Sankarapu. “SafeTune: A Unified Faithful Library for Auditing and Repairing Safety Drift in Fine-Tuned LLMs”. In *Conference on Empirical Methods in Natural Language Processing (EMNLP), Demo Track*. 2026.

*Equal contribution / co-first authorship.

WORKSHOP PROCEEDINGS

- [W1] **Saisab Sadhu**, Biswajit Patra, and Tanmay Basu. “[Structured Adversarial Synthesis: A Multi-Agent Framework for Generating Persuasive Financial Analysis from Earnings Call Transcripts](#)”. In *Workshop on Financial Technology and Natural Language Processing (FinNLP)*, EMNLP. Nov. 2025.
- [W2] **Saisab Sadhu**, Shreyans Arora, and Pratinav Seth. “Cited but Not Consulted: A Counterfactual Audit of Legal Chain-of-Thought Faithfulness”. In *Workshop on Natural Legal Language Processing (NLLP)*, EMNLP. 2026.
- [W3] **Saisab Sadhu**, Aadit Sengupta, Vinay Kumar Sankarapu, and Pratinav Seth. “Verdict Without the Rule: Diagnosing and Auditing Regulatory Rule Sensitivity in LLM Compliance Systems”. In *Workshop on Natural Legal Language Processing (NLLP)*, EMNLP. 2026.

TECHNICAL SKILLS

Languages	Python, C, R, SQL, Matlab, Bash, PySpark
ML/DL & Data	PyTorch, TensorFlow, Keras, Sk-learn, Hugging Face (Transformers, PEFT), pandas, NumPy
NLP & GenAI	NLTK, Gensim, spaCy, stanza, AutoGen, DPO, LoRA/QLoRA, RLHF, G-Eval
Dev & Tooling	Git, Docker, Kubernetes, SLURM, Databricks, Weights & Biases, BeautifulSoup, Selenium

ACHIEVEMENTS

- **Best Paper Award, SIGIR 2026:** Received the Best Paper Award at the 49th International ACM SIGIR Conference for “*When RAG Disagrees: Detecting Latent Epistemic Conflict via Logit Interactions*”.
- **AAAI 2026 EGSAI Selection:** Selected to present “*Hierarchical Pedagogical Oversight: A Multi-Agent Adversarial Framework for Reliable AI Tutoring*” at AAAI 2026; one of 51 works chosen for global submissions.
- **Top Performer, FinNLP @ EMNLP 2025:** Ranked first globally on the official ‘Win Rate vs Analyst Report’ metric (Overall 3rd); system reports were preferred over professional human analysts.
- **Recipient, Student Innovation Grant (Rs. 2 Lakhs):** Awarded by IICE (Funded by DST, GOI) to develop an AI fintech platform; demonstrated 68% profit increase in backtesting.
- **CARE Conference Travel Grant:** Awarded full registration waiver and travel support (IIT Guwahati) for poster presentation at the Collaborative for Academic Research Excellence Conference.

POSITION OF RESPONSIBILITY

- **Reviewing & Program Committee Service:** 2026 – 2027
Program Committee Member, AI Alignment Track (AAAI-27); peer reviewer for NeurIPS 2026, workshops co-located with (EMNLP 2026, NeurIPS 2026, ICML 2026), and the Global South in AI across venues.
- **Secretary, Student Development Council & Founded, Entrepreneurship Cell - IISERB:** 2024
Organized institute-wide initiatives, workshops, and panel discussions to promote student growth and entrepreneurship.
- **Institute Placement Coordinator, Center for Career Development:** Aug 2022 – Oct 2023
Coordinated campus placements and industry outreach; achieved a notable increase in recruiter engagement.

REFERENCES

- Dr. Tanmay Basu** Assistant Professor & HOD, Dept. of Data Science and Engineering, IISER Bhopal
- Dr. Dwaipayan Roy** Assistant Professor, Dept. of Computational and Data Sciences, IISER Kolkata

UNDER REVIEW

- [C4] **Saisab Sadhu**, Aadit Sengupta, Vinay Kumar Sankarapu, and Pratinav Seth. “What Do Compliance Detectors Read? An Audit of Activation Probes and Guard Models”. *Under Review*.
- [C5] **Saisab Sadhu**, Pratinav Seth, and Vinay Kumar Sankarapu. “Forgetting That Sticks: Quantization-Permanent Unlearning via Circuit Attribution”. *Under Review*.
- [C6] Pratinav Seth*, Anshul Kaushal*, **Saisab Sadhu***, and Vinay Kumar Sankarapu. “Drift, Then Repair: A Controlled Cross-Paradigm Audit of Safety in Fine-Tuned LLMs”. *Under Review, Alignment Track*.
- [C7] **Saisab Sadhu** and Pratinav Seth. “The Mechanistic Flinch: Predicting Multi-Turn Sycophantic Collapse from First-Contact Attention Dynamics”. *Under Review, Alignment Track*.

*Equal contribution / co-first authorship.